EXPERIMENT 10: Install Google App Engine SDK on Windows and Run a Hello World Application

1. Aim

To install the Google Cloud development tools on Microsoft Windows and create and execute a simple Hello World application using Google App Engine.

2. Procedure

1. Open a web browser and visit the Google Cloud website.
2. Download the Google Cloud CLI installer for Windows.
3. Run the installer and complete the installation steps.
4. Install Python if it is not already available on the system.
5. Open the Google Cloud SDK Shell or Command Prompt.
6. Initialize the CLI using gcloud init.
7. Sign in using a Google account and select the required Google Cloud project.
8. Create a directory named HelloWorld.
9. Inside the directory, create a Python file named main.py.
10. Enter the Flask program that displays the Hello World message.
11. Create requirements.txt containing Flask.
12. Create app.yaml and specify the supported App Engine runtime.
13. Deploy the application using gcloud app deploy.
14. After deployment, run gcloud app browse and verify the message in the browser.

3. Program

File – main.py:

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "<h1>Hello World!</h1>"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=8080)
```

File – requirements.txt:

```
Flask
```

File – app.yaml:

```
runtime: python313
```

Deployment commands:

```
gcloud init
gcloud app deploy
gcloud app browse
```

4. Output of Program

```
Browser output:

Hello World!
```

```
Application deployed successfully.
```

5. Result

Thus, the Google Cloud development tools were installed on Windows, and a simple Hello World application was created and deployed using Google App Engine.

EXPERIMENT 10: Install Google App Engine SDK on Windows and Run a Hello World Application

1. Aim

To install the Google Cloud development tools on Microsoft Windows, create a simple Python-based Hello World web application using Flask and deploy the application using Google App Engine.

2. Procedure

1. Start the Windows operating system and open a web browser. 2. Navigate to the official Google Cloud website and locate the Google Cloud CLI download section. 3. Download the Google Cloud CLI installer suitable for Windows. 4. Execute the downloaded installer and follow the installation wizard. 5. Install Python if it is not already available on the system. 6. Open Command Prompt or Google Cloud SDK Shell. 7. Verify the installation by executing gcloud --version. 8. Initialize the Google Cloud CLI using the command gcloud init. 9. Sign in using a valid Google account and select the required Google Cloud project. 10. Enable billing if required by the selected App Engine configuration. 11. Create a new project directory named HelloWorld. 12. Inside the directory, create a Python file named main.py. 13. Import the Flask module and create a Flask application instance. 14. Define a route for the home page that returns the message Hello World!. 15. Create a file named requirements.txt and include Flask as the application dependency. 16. Create an app.yaml file and specify the supported Python runtime. 17. Save all the application files in the same project directory. 18. Open the command prompt inside the project directory. 19. Execute gcloud app deploy to upload and deploy the application to Google App Engine. 20. Wait for the deployment process to finish and check for successful deployment. 21. Execute gcloud app browse to open the deployed application in a web browser. 22. Verify that the browser displays the Hello World message.

4. Output of Program

Expected Browser Output: Hello World! Expected Result: Application deployed successfully. Explanation: The Flask application handles requests made to the home page and returns a simple HTML response containing the Hello World message. Google App Engine hosts the application and makes it accessible through a web URL.

5. Result

Thus, the Google Cloud CLI was successfully installed and configured on Windows. A simple Hello World application was developed using Python and Flask, deployed on Google App Engine and accessed successfully through a web browser.